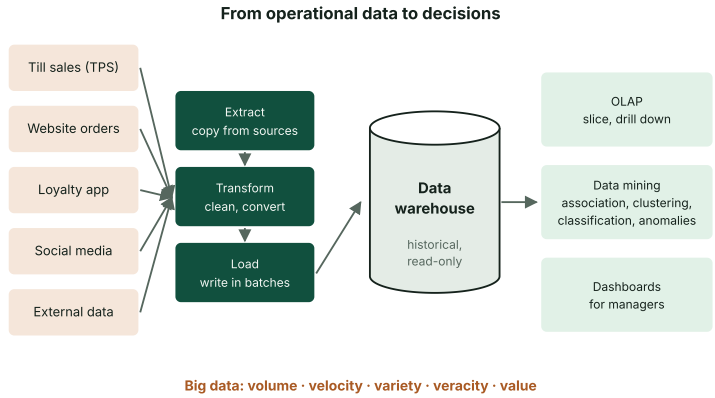
Big data, data warehousing, data mining and analytics

IS08 FACT FILE

Specification: students should be able to

- Describe the characteristics of big data
- Explain data warehousing, including extract–transform–load
- Describe data mining techniques and the use of analytics in organisations
- Discuss privacy issues arising from big data and data mining

Key words: big data, volume, velocity, variety, veracity, data warehouse, ETL, OLAP, data mining, association, clustering, predictive analytics



Data from operational sources is extracted, transformed and loaded into a data warehouse, then analysed with OLAP and data mining to support decisions

Big data

Big data describes data sets whose volume, velocity and variety make them too large or complex for traditional database tools. Veracity (accuracy and trustworthiness) and value (the useful insight that can be extracted) are often added. Much big data is unstructured, such as text, images and sensor streams, so it is stored and processed across clusters of servers in parallel, often in NoSQL databases rather than relational tables.

Data warehouses

A data warehouse is a central repository of integrated, historical data drawn from many operational systems. Data is extracted, transformed into a consistent and clean format, and loaded (ETL). Warehouses are subject-oriented, time-variant and non-volatile, and are usually denormalised to make complex analytical queries faster. Managers use OLAP tools to slice data and drill down, for example from yearly sales to monthly sales by branch.

Data mining and analytics

Data mining uses statistical and machine learning techniques to discover hidden patterns in large data sets. Common techniques are association rules, classification, clustering, anomaly detection and regression. Results support descriptive, predictive and prescriptive analytics, helping organisations target marketing, manage stock, detect fraud and plan services. The benefits must be weighed against cost, data quality and the privacy of the individuals whose data is mined.

T Key terms from the mark schemes

Big data
Data sets so large, fast-moving or varied that they cannot be processed efficiently with traditional database tools.

ETL
Extract, transform, load: the process of copying data from source systems, cleaning and converting it, and loading it into a data warehouse.

OLAP
Online analytical processing: tools that let users analyse multidimensional data, for example by slicing and drilling down.

Data warehouse
A large store of integrated, historical, read-only data from many sources, organised for analysis and decision-making.

Data mining
The process of analysing large data sets to discover previously unknown patterns, relationships and trends.

Exam focus

Extended questions on big data and data mining are often marked with levels of response including quality of written communication. Define the terms precisely, apply them to the scenario, and balance benefits against issues such as privacy, consent, security and bias, linking to the Data Protection Act 2018 and UK GDPR where relevant. Avoid listing techniques without saying what each finds. A clear conclusion that weighs the points is needed for the top band.

? Check yourself

1. Describe three characteristics of big data. [3]
2. Explain the stages of ETL used when loading a data warehouse. [3]
3. Give two differences between an operational database and a data warehouse. [2]

Answers: 1. Any three from, each described: Volume – huge quantities of data, terabytes to petabytes (1) / Velocity – data arrives very quickly and must be processed fast, often in real time (1) / Variety – many types and formats, structured and unstructured such as text, images and video (1) / Veracity – data may be of uncertain accuracy or trustworthiness (1) / Value – useful insight can be gained from it (1). 2. Extract: data is copied from the various operational source systems (1) / Transform: data is cleaned and made consistent, e.g. converting formats, removing duplicates and errors (1) / Load: the transformed data is loaded into the data warehouse for analysis (1). 3. Any two from: operational databases support daily transactions, a warehouse supports analysis and decision-making (1) / operational data is current and changing, warehouse data is historical and read-only (non-volatile) (1) / operational databases are normalised to 3NF, warehouses are often denormalised (e.g. star schema) (1) / operational queries are many short updates, warehouse queries are few, large, complex reads (1) / operational users are staff, warehouse users are managers/analysts (1) / warehouse integrates data from many sources (1).